Towards the Erdős-Szemerédi Sum-Product Conjecture

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July 31, 2026

Abstract

This document outlines a rigorous approach towards resolving the Erdős-Szemerédi Sum-Product Conjecture. We establish foundational lemmas regarding the additive and multiplicative energies of finite subsets of natural numbers.

1 Introduction

The Erdős-Szemerédi Conjecture postulates that for any finite set $A \subset \mathbb{N}$, either its sum set $A + A$ or its product set $A \cdot A$ must be large.

Definition 1.1. *Let $A \subset \mathbb{N}$ be a finite set. The sum set is defined as $A + A = \{a + b \mid a, b \in A\}$. The product set is defined as $A \cdot A = \{a \cdot b \mid a, b \in A\}$.*

Conjecture 1.2 (Erdős-Szemerédi, 1983)**.** *For every $\varepsilon > 0$, there exists a constant $c > 0$ such that for all finite sets $A \subset \mathbb{N}$,*

$$\max(|A + A|, |A \cdot A|) \geq c|A|^{2-\varepsilon}$$

2 Foundational Lemmas

We begin by establishing a relationship based on additive energy.

Definition 2.1. *The additive energy of a set A, denoted $E^+(A)$, is the number of quadruples $(a, b, c, d) \in A^4$ such that $a + b = c + d$.*

Lemma 2.2. *For any finite set $A \subset \mathbb{N}$, $E^+(A) \geq \frac{|A|^4}{|A+A|}$.*

Proof. Let $r_{A+A}(x)$ be the number of pairs $(a, b) \in A \times A$ such that $a + b = x$. By definition, $\sum_{x \in A+A} r_{A+A}(x) = |A|^2$. Furthermore, the additive energy can be expressed as $E^+(A) = \sum_{x \in A+A} r_{A+A}(x)^2$. Applying the Cauchy-Schwarz inequality, we obtain:

$$\left(\sum_{x \in A+A} r_{A+A}(x) \right)^2 \leq |A + A| \sum_{x \in A+A} r_{A+A}(x)^2$$

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Substituting the sums, we get:

$$(|A|^2)^2 \leq |A+A| E^+(A)$$

Rearranging the terms yields:

$$E^+(A) \geq \frac{|A|^4}{|A+A|}$$

$\square$